

FIFO pointers

A FIFO buffer decouples producer and consumer with first-in first-out order, the oldest entry is read next

Push A5 starter

- Starter: depth-four empty FIFO
- Push hex A5
- Count becomes one, empty clears, write pointer advances to one, read pointer stays at zero
- Data A5 sits in slot zero
- Pop once returns A5, count returns to zero, read pointer advances to one
- Fill with four pushes to set full; a fifth push logs blocked FULL

Browser lab

Digital Design and Verification Platform

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INTERACTIVE TOOL

FIFO pointer & flags

Step a small synchronous FIFO: write/read pointers, occupancy count, and empty / full flags. Push and pop to see the circular buffer move. Starter: depth-4 empty FIFO — push 0xA5.

Starter example: depth-4 empty FIFO. Push 0xA5 — count becomes 1, empty clears, wr advances.

Challenge 0 / 22 cleared

Quiz: FIFO: FIFO means...

☐ First-In First-Out — oldest entry is read next

☐ First-In Last-Out (a stack)

☐ random access like RAM

☐ only a clock generator

Show hint

Check

Next

Load challenge setup

Idle

Quiz: FIFO

Quiz: empty

Quiz: full

Quiz: write ptr

Quiz: read ptr

Quiz: push when full

Quiz: pop when empty

Quiz: sync FIFO

Push 0xA5

Fill to full

Push blocked

Pop recovers A5

Pop blocked

Write wrap

FIFO order

FIFO lab starter

Workbook practice

- On paper, draw a depth-four FIFO with wr and rd pointers
- Trace push A5, then pop A5, listing count and pointers after each step
- Show what happens when wr wraps after four pushes
- Tabulate empty versus full conditions
- Name one pitfall: popping when empty or pushing when full

Pitfalls to watch

- Do not confuse `wr equals rd` with empty
- Some pointer-only FIFOs leave one slot spare to detect full
- Async clock domains need Gray-coded pointers and synchronizers, beyond this sync lab
- And remember

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, push A5, pop it back, then fill and drain once
- On paper, trace FIFO order for two pushes and one pop
- When you are ready, take the short quiz, then continue to cache walk